Impact of Climate Change: Trend & Correlation Analysis of Temperature, Precipitation, and Vector-Borne Diseases

Monroe County (2002-2024) Data

Correlation: Malaria, Precipitation, and Temperature

	Precipitation	Temperature	Malaria
Precipitation	1.000	-0.238	-0.277
Temperature	-0.238	1.000	0.507
Malaria	-0.277	0.507	1.000

Table: Monroe County 2002-2024(Q3) Data

- **Temperature and Malaria:** Moderate positive correlation ($r = 0.507$).
- **Precipitation and Malaria:** Weak negative correlation ($r = -0.277$).

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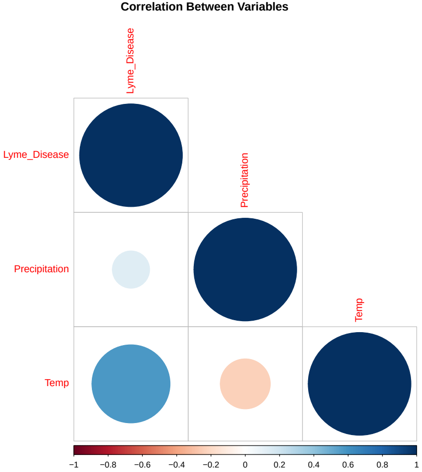
Implications

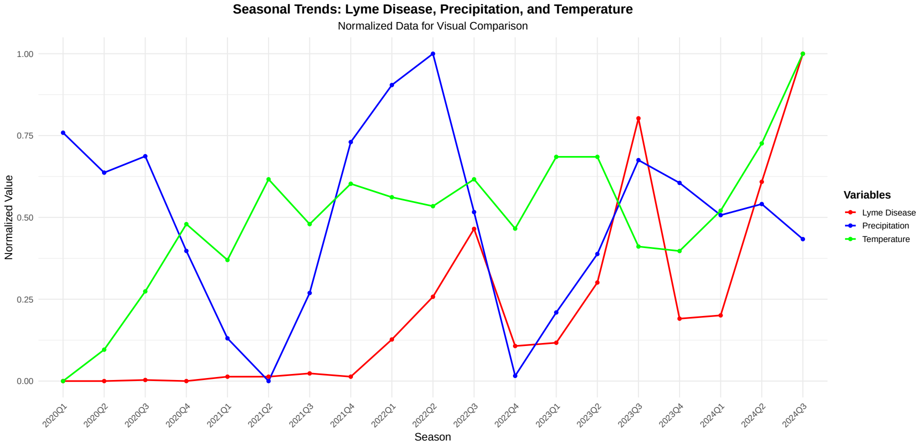
- Higher temperatures may be associated with increased malaria cases.
- Precipitation shows a weaker influence.

Correlation: Lyme Disease, Precipitation, and Temperature



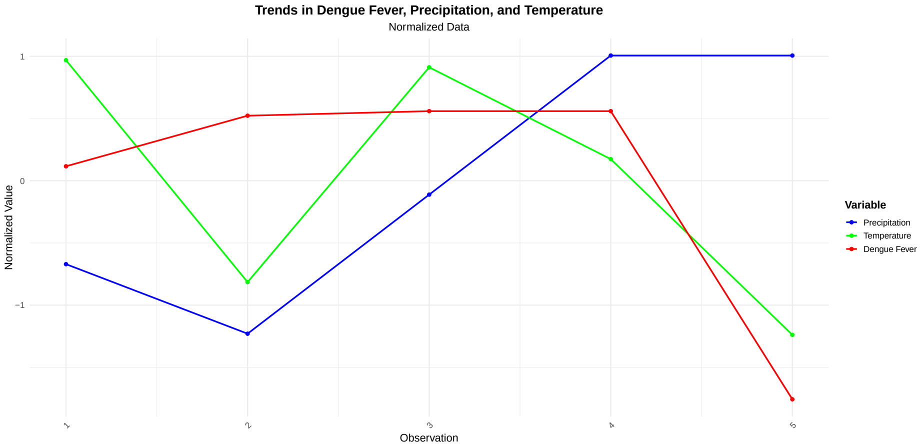
- **Lyme Disease and Temperature:** Moderate positive correlation ($r = 0.578$).
- **Lyme Disease and Precipitation:** Weak positive correlation ($r = 0.132$).
- **Precipitation and Temperature:** Weak negative correlation ($r = -0.238$).

Lyme Disease and Climate Change Trend



2020-2024(Q3) Monroe County Data

Lyme Disease and Climate Change Trend



2020-2024(Q3) Monroe County Data

Descriptive Statistics of Disease Incidence (2020-2024)

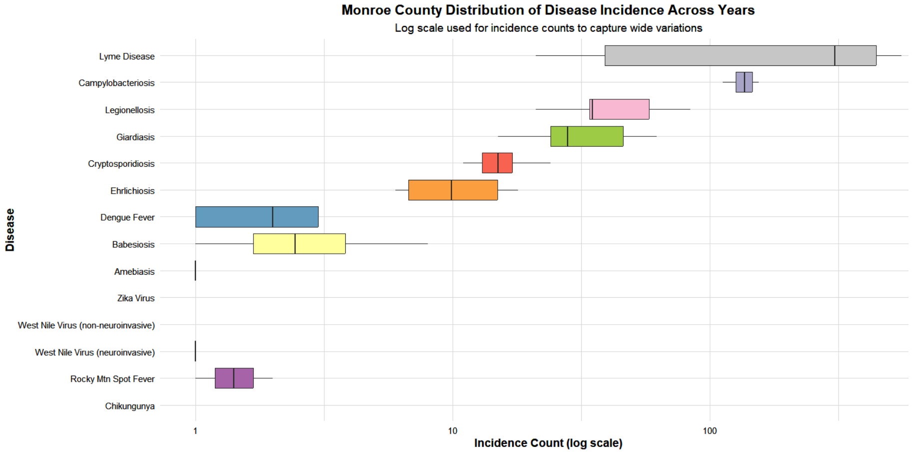
Disease	Slope	p-value	Mean	SD	Interpretation
Amebiasis	2×10^{-1}	0.182	0.8	0.45	Slight positive trend; not statistically significant.
Babesiosis	8×10^{-1}	0.497	2.8	3.11	Moderate positive trend; not statistically significant.
Campylobacteriosis	6.8	0.246	135	16.82	Positive trend; not statistically significant.
Chikungunya	0	NaN	0	0	No reported cases; trend analysis is not applicable.
Cryptosporidiosis	1.1	0.566	16	5	Slight positive trend; not statistically significant.
Dengue Fever	3×10^{-1}	0.420	2	1	Slight positive trend; not statistically significant.
Ehrlichiosis	3.9	0.0539	9	7.07	Positive trend; marginal statistical significance (close to $p = 0.05$).
Giardiasis	9	0.140	35	18.84	Positive trend; not statistically significant.
Legionellosis	-2.16×10^{-13}	1.000	46.4	24.89	No trend; entirely flat across years.
Lyme Disease	147	0.00458	272.8	238.75	Strong positive trend; statistically significant ($p \leq 0.01$).
Rocky Mtn Spot Fever	2×10^{-1}	0.559	0.6	0.89	Slight positive trend; not statistically significant.
West Nile Virus (neuroinvasive)	1×10^{-1}	0.559	0.2	0.45	Slight positive trend; not statistically significant.
West Nile Virus (non-neuroinvasive)	0	NaN	0	0	No reported cases; trend analysis is not applicable.
Zika Virus	0	NaN	0	0	No reported cases; trend analysis is not applicable.

Table: Descriptive Statistics of Monroe County Selected Disease Incidence

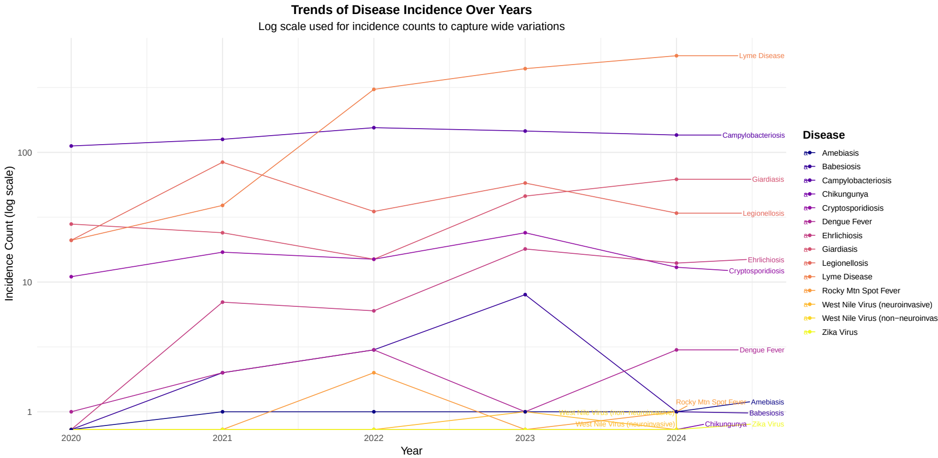
Interpretation Highlights

- Lyme Disease and Ehrlichiosis show strongest positive trends and statistical significance among selected diseases.
- The following have meaningful positive trend of increase in years; Amebiasis, Babesiosis, Campylobacteriosis, Cryptosporidiosis, Dengue Fever, Giardiasis, Rocky Mtn Spot Fever and West Nile Virus (neuroinvasive).
- Diseases with meaningful positive trend, while not statistically significant, may require close monitoring, as they exhibit a marginally significant upward trend.

Distribution of selected Disease Incidence Across Years



Trends of selected Disease Incidence Across Years



Prevalence Trends of selected Disease Across Years

The formula for Prevalence is:

$$\text{Prevalence for Year} = \left(\frac{\text{Incidence in Year}}{\text{Total Incidence (2020-2024)}} \right) \times 100$$

Prevalence of Disease Incidence Over Years (2020-2024)

Disease	2020	2021	2022	2023	2024
Babesiosis	0.00	5.56	8.33	22.22	2.78
Dengue Fever	6.67	13.33	20.00	6.67	20.00
Ehrlichiosis	0.00	10.00	8.00	22.22	17.78
Lyme Disease	1.12	2.13	16.51	23.73	29.47
Rocky Mtn Spot Fever	0.00	0.00	5.88	0.00	2.22
West Nile Virus (neuroinvasive)	0.00	0.00	0.00	100.00	0.00
Legionellosis	9.52	38.10	15.60	24.67	14.79
Amebiasis	0.00	3.33	3.33	3.33	3.33
Giardiasis	9.60	8.00	5.33	16.00	21.33
Cryptosporidiosis	8.33	12.50	10.42	15.00	8.33
Campylobacteriosis	16.15	18.52	22.60	21.92	20.11

Table: Prevalence of Disease Incidence Over Years (2020-2024)

Prevalence and Public Health Implications (2020–2024)

- Tickborne Diseases:
  - Lyme Disease: Steady rise, reaching 29.47% in 2024.
  - Ehrlichiosis: Increasing trend, peaking in 2023 (22.22%).
  - Babesiosis: Peaked in 2023 (22.22%) but sharply declined in 2024 (2.78%).

  - **Implication**: Focus on habitat management, public awareness, and early diagnosis.

- Vectorborne Diseases:
  - Dengue Fever: Fluctuated with peaks in 2022 (20%) and 2024 (20%).
  - West Nile Virus (neuroinvasive): Surged in 2023 (100%) but absent in other years.

  - **Implication**: Strengthen mosquito control during outbreak years.

Prevalence and Public Health Implications (2020–2024)

- Waterborne Diseases:
  - Legionellosis: Peaked in 2021 (38.10%), remained significant throughout.
  - Cryptosporidiosis and Giardiasis: Fluctuations indicate potential waterborne outbreaks.

  - **Implication**: Improve water supply monitoring and public education.

- Actionable Recommendations:
  - Enhance vector and tick control measures during peak seasons.
  - Strengthen food and water safety programs.
  - Raise public awareness about preventive strategies.

Reference

- Data Source from Monroe County, Department of Epidemiology and Disease Control(2024)